

Choline supplementation in early life improves and low levels of choline can impair outcomes in a mouse model of Alzheimer's disease

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Abstract

Maternal choline supplementation (MCS) improves cognition in Alzheimer's disease (AD) models. However, effects of MCS on neuronal hyperexcitability in AD are unknown. We investigated effects of MCS in a well-established mouse model of AD with hyperexcitability, the Tg2576 mouse. The most common type of hyperexcitability in Tg2576 mice, and many other mouse models and AD patients, are generalized EEG spikes (interictal spikes; IIS). Hyperexcitability is also reflected by elevated expression of the transcription factor Δ FosB in the principal cells of the dentate gyrus (DG), granule cells (GCs). We also studied the hilus of the DG because hilar neurons regulate GC excitability. We found reduced expression of the neuronal marker NeuN within hilar neurons in Tg2576 mice, which other studies have shown is a sign of oxidative stress or other pathology.

Tg2576 breeding pairs received a diet with a relatively low, intermediate or high concentration of choline. After weaning, all mice received the intermediate diet. In offspring of mice fed the high choline diet, IIS frequency declined, GC Δ FosB expression was reduced, and NeuN expression restored. Spatial memory improved. In contrast, offspring exposed to the relatively low choline diet had several adverse effects, such as increased mortality. The results provide new evidence that a diet high in choline in early life can improve outcomes in a mouse model of AD, and relatively low choline can have negative consequences. This is the first study showing that dietary choline can regulate hyperexcitability, hilar neurons, and spatial memory in an animal model of AD.

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In this **fundamental** work, the authors demonstrated that maternal choline supplementation improved spatial memory, reduced hyperexcitability, and restored NeuN expression in a familial Alzheimer's disease mouse model. Interestingly, choline deficiency increased mortality, while paradoxically reduced hyperexcitability. Using behavior, electrophysiological, and histological measures, the authors present **solid** and **convincing** evidence supporting the significant role of maternal choline supplementation in protecting hippocampal functions vulnerable to Alzheimer's disease.

I. INTRODUCTION

Diet has been suggested to influence several aspects of brain health. One dietary intervention that has been studied extensively in rodents and humans is supplementation of the maternal diet with the nutrient choline, (maternal choline supplementation; MCS). MCS improves many aspects of brain health in humans (Zeisel & da Costa 2009 [\[1\]](#), Jiang et al. 2014 [\[2\]](#)). In normal rats, MCS also is beneficial, with numerous studies showing improved behavior in the adult offspring (for review, see (for review see Meck & Williams 2003 [\[3\]](#))).

In Alzheimer's disease (AD), changes to the diet have been recommended (Bourre 2006 [\[4\]](#), Power et al. 2019 [\[5\]](#), Mao et al. 2021 [\[6\]](#), Lobo et al. 2022 [\[7\]](#)), including MCS (Strupp et al. 2016 [\[8\]](#), Velazquez et al. 2020 [\[9\]](#)). The recommendation to use MCS is based primarily on data from the Ts65Dn mouse model of Down syndrome (DS) and AD, where MCS led to improved memory and attention in Ts65Dn offspring (Strupp et al. 2016 [\[8\]](#), Powers et al. 2017 [\[10\]](#), Powers et al. 2021 [\[11\]](#)). In addition, degeneration of basal forebrain cholinergic neurons (BFCNs) in Ts65Dn mice, a hallmark of DS and AD, was reduced (Kelley et al. 2016 [\[12\]](#), Powers et al. 2017 [\[10\]](#)).

Based on the studies of Ts65Dn mice, we asked if MCS would improve another impairment that has been found in AD patients and mouse models of AD, hyperexcitability. We chose to study Tg2576 mice because hyperexcitability is robust, and BFCNs are likely to play a role (Kam et al. 2016 [\[13\]](#), Lisgaras & Scharfman 2023 [\[14\]](#)). In our past work, the primary type of hyperexcitability was interictal spikes (IIS), which are named because they occur in between seizures (ictal events) in epilepsy. They are studied by EEG. We also studied hyperexcitability using Δ FosB expression, a transcription factor that is increased when neurons have been highly active over the prior 10-14 days (McClung et al. 2004 [\[15\]](#)). We studied Δ FosB in granule cells (GCs) of the dentate gyrus (DG), because numerous Δ FosB-expressing GCs occur when there are IIS (You et al. 2017 [\[16\]](#), You et al. 2018 [\[17\]](#)).

We also asked if behavior would improve if MCS reduced IIS frequency in Tg2576 mice. The basis for this question is in studies of epilepsy, where IIS disrupt cognition (Rausch et al. 1978 [\[18\]](#), Aarts et al. 1984 [\[19\]](#), Holmes & Lenck-Santini 2006 [\[20\]](#), Kleen et al. 2010 [\[21\]](#), Kleen et al. 2013 [\[22\]](#), Gelinias et al. 2016 [\[23\]](#)). Also, IIS in the Tg2576 mouse start early in life (by 1 month of age), before memory impairment, which occurs at 3-4 months of age (Duffy et al. 2015 [\[24\]](#)). Therefore, IIS might disrupt activity sufficiently to promote behavioral impairments. We previously found that the novel object location (NOL) task is impaired at 3-4 months of age in Tg2576 mice (Duffy et al. 2015 [\[24\]](#)), so we evaluated NOL, and a related task, novel object recognition (NOR; Vogel-Ciernia & Wood 2014 [\[25\]](#)). We also studied expression of NeuN, a neuronal marker that is reduced in numerous disease conditions (Buckingham et al. 2008 [\[26\]](#), Kadriu et al. 2009 [\[27\]](#), Matsuda et al. 2009 [\[28\]](#), Duffy et al.

2011 [↗](#), Duffy et al. 2015 [↗](#)). Here we report loss of NeuN-ir in the DG hilus of Tg2576 mice, which is important because impaired hilar neurons could cause GC hyperexcitability (Sperk et al. 2007 [↗](#), Scharfman & Myers 2012 [↗](#)).

We fed dams one of three diets, which were relatively low, intermediate or high in choline. The high choline diet provided levels of choline similar to other studies of MCS in rodents (Meck et al. 1988 [↗](#), Loy et al. 1991 [↗](#), Holler et al. 1996 [↗](#), Meck & Williams 1999 [↗](#), Sandstrom et al. 2002 [↗](#), Mellott et al. 2004 [↗](#), Glenn et al. 2012 [↗](#), Kelley et al. 2019 [↗](#)). After weaning the intermediate diet was used. Offspring were implanted with electrodes for EEG at 1 month of age, and 24 hr-long recordings began 1/month starting at 1 week after surgery. NOL and NOR were tested at 3 months and again at 6 months, and then we perfusion-fixed mice, sectioned the brains, and evaluated Δ FosB and NeuN protein expression (**Figure 1A** [↗](#)).

The results showed a remarkable effect of the high choline diet. IIS and spatial memory were improved, as was Δ FosB and NeuN expression. Interestingly, the relatively low choline diet had mixed effects, reducing IIS frequency, but making memory, Δ FosB, and NeuN worse. The mice also died prematurely. In summary, we make a strong argument for choline supplementation in early life to improve outcomes in an AD model, and potentially AD.

II. METHODS

A. Animals

All experimental procedures followed the guidelines set by the National Institute of Health (NIH) and were approved by the Institutional Animal Care and Use Committee (IACUC) of the Nathan Kline Institute for Psychiatric Research. Mice were housed in standard mouse cages with corn cob bedding and a 12-hr light-dark cycle. Food and water were provided *ad libitum*.

Tg2576 mice express a mutant form of human APP (isoform 695) with a mutation found in a Swedish cohort (Lys670Arg, Met671Leu). The expression is driven by the hamster prion protein promoter. Mice were bred in-house from heterozygous Tg2576 males and non-transgenic female mice (C57BL6/SJL F1 hybrid). Wildtype (WT) mice were obtained from Jackson Laboratories (Stock# 100012). At weaning, genotypes were determined using a protocol to detect APP695.

Breeding pairs were randomly assigned to receive one of three diets with different concentrations of choline chloride: 1.1 g/kg (AIN-76A, Dyets), 2.0 g/kg (Purina 5008, W.F. Fisher and Son Inc.) and 5.0 g/kg (AIN-76A, Dyets; **Supplemental Table 1** [↗](#)). These diets (relatively “low choline” diet, “intermediate choline” diet and “high choline” diet, respectively) were used until weaning (25-30 days of age). After weaning, all mice were fed the intermediate diet. Mice were housed with others of the same sex (1-4 per cage).

B. Behavior

1. General information

Mice were housed in the room where they would be tested behaviorally starting at least 24 hr prior to all experiments. Both NOL and NOR included three acclimations (5 min each) followed by a training and a testing session (5 min each; **Figure 1B-D** [↗](#)). The interval between training and testing was 60 minutes and therefore tested short-term memory (Vogel-Ciernia & Wood 2014 [↗](#)). NOL and NOR tests were separated by 7 days to ensure effects of one task did not affect the next. Prior studies have failed to find an effect on one task on the next when a 7 day-long interval is used (Botterill et al. 2021 [↗](#)). The order of testing (NOL before NOR or NOR before NOL) was randomized.

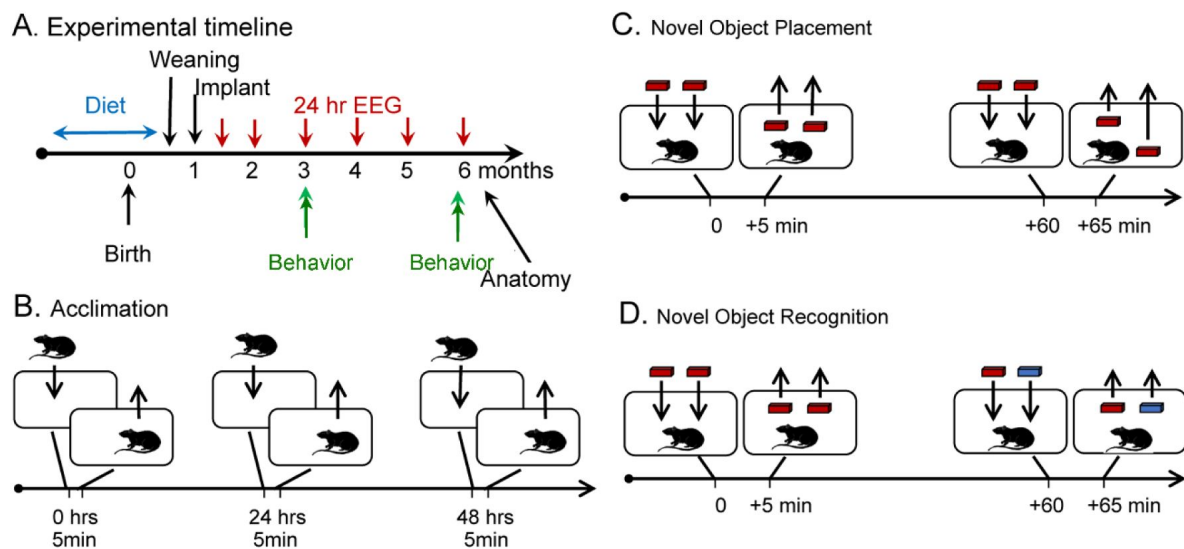


Figure 1.

Schematic of the behavioral procedures for the NOL and NOR tasks.

A. Experimental timeline. Dams were fed one of three diets for mating, gestation, and offspring consumed the diet until weaning at 25-30 days of age. At 1 month-old, offspring were implanted with electrodes for EEG and recorded for a 24 hr period at 1, 2, 3, 4, 5, and 6 months of age. At 3 and 6 months of age behavior was tested. At 6 months of age, after behavioral testing, mice were perfused, brains were sectioned, and sections were processed with antibodies against NeuN and Δ FosB.

B. Prior to NOL or NOR, animals were acclimated to the testing arena. There were three acclimation sessions separated by 24hrs during which animals were allowed to freely explore for 5 min.

C. In the NOL task, animals were placed in a cage with two identical objects and allowed to freely explore for 5 min (Training). After one hour they were brought back to the cage, where one object was displaced, and allowed to freely explore for 5 min (Testing).

D. In the NOR task, animals were placed in a cage with two identical objects and allowed to freely explore for 5 min (Training). After one hour they were brought back to the cage, where one object was replaced, and allowed to freely explore for 5 min (Testing).

Video recordings of all training and testing sessions were captured using a USB camera (Logitech HD Pro C920, Logitech). Acclimations were conducted between 10:00 a.m. and 12:00 p.m., while training and testing was conducted between 1:00 p.m. and 4:00 p.m. All equipment was cleaned using 70% ethanol before each use. In pilot studies, we chose objects that would be explored equally by animals. Indeed, animals showed no preference for either of the objects (i.e., LEGO object and pineapple, **Supplemental Figure 1** [↗](#)) during training, as reported in the Results.

2. NOL

NOL was conducted in a standard rat cage (26 cm wide x 40 cm long x 20 cm high) with different pictures on 3 sides of the cage to provide a consistent context and therefore foster acclimation. Pictures included several shapes and colors. The dimensions of the pictures were 1) 10 cm x 21 cm, 2) 16 cm x 18 cm, and 3) 17 cm x 20 cm. During the acclimations, animals were allowed to freely explore the cage. During the training session, mice were placed in the cage where they had been acclimated, with two identical objects, 1 in a corner of the cage (e.g., left top) and the second in the adjacent corner (e.g., left bottom, **Supplemental Figure 1** [↗](#)). The mice were then removed and placed in their home cage for 1 hr. During the test session, one of the objects was moved to the opposite end of the cage (left top to right top; **Supplemental Figure 1** [↗](#)). The two objects were identical pineapple-like objects made of metal that were painted and approximately 3 cm x 3 cm x 5 cm (**Supplemental Figure 1** [↗](#); Botterill et al. 2021 [↗](#)).

3. NOR

NOR was conducted in a standard rat cage as described above. During the acclimations, animals were allowed to freely explore the cage. During the training session, mice were placed in the cage used for acclimation with two identical objects centered along the shortest cage wall (**Supplemental Figure 1** [↗](#)). The mouse was then removed and placed in their home cage for 1 hr. During the testing session, one of the objects was replaced with a different one. The objects that were identical were 2 pineapple-like objects, and the new object was made of 2 green LEGO pieces approximately 5 cm x 7 cm x 7 cm (**Supplemental Fig. 1** [↗](#); Botterill et al. 2021 [↗](#)).

4. Quantification

The experimenter who conducted the analysis was blind to the genotype of the offspring and which object was novel or familiar. Videos of the training and testing sessions were analyzed offline. Exploration was quantified based on the guidelines of Vogel-Ciernia & Wood (2014) [↗](#), but in addition to time spent exploring, the number of approaches to an object was also quantified. Time spent exploring each object was defined as the duration of time the nose was pointed at the object and the nose was located within 2 cm of the object. In addition, exploration time included the time animals spent on top of the object if they were looking down and sniffing it. An approach was defined by a movement toward the object and ending at least 2 cm from the edge of the object.

Animals that remembered the objects explored during the training phase were expected to demonstrate an increased preference for novelty in the test phase (Ennaceur & Delacour 1988 [↗](#)). In other words, exploration of the novel object during testing was expected to be higher than 50% of the total time of object exploration.

C. Anatomy

1. Perfusion-fixation and sectioning

Mice were initially anesthetized by isoflurane inhalation (NDC# 07-893-1389, Patterson Veterinary), followed by urethane (2.5 g/kg; i.p.; Cat# U2500, Sigma-Aldrich). Under deep anesthesia, the abdominal and heart cavities were opened and the animal was transcardially-perfused using 10 mL of room temperature (RT) saline (0.9% sodium chloride in double distilled

(dd) H₂O) using a peristaltic pump (Minipuls1; Gilson), followed by 30 mL of 4°C 4% paraformaldehyde (PFA, Cat# 19210, Electron Microscopy Sciences) in 0.1 M phosphate buffer (PB; pH 7.4). The brains were immediately extracted, hemisected, and post-fixed for at least 24 hr in 4% PFA (at 4°C).

Following post-fixation, the left hemisphere was sectioned in the coronal plane and the right hemisphere was sectioned in the horizontal plane (50 µm-thick sections) using a vibratome (VT 1000P, Leica Biosystems). Sections were collected serially and stored at 4°C in 24-well tissue culture plates containing cryoprotectant solution (30% glycerol, 30% ethylene glycol, 40% 0.1 M phosphate buffer, pH 6.7). Throughout the hippocampus, every 6th section was processed with an antibody to NeuN or ΔFosB.

2. NeuN and ΔFosB immunohistochemistry

Free floating sections were washed in 0.1 M Tris Buffer (TB; 3 x 5 min), followed by a 3 min wash in 1% H₂O₂. Sections were then washed in 0.1 M TB (3 x 5 min) and incubated for 60 min in 5% horse serum for NeuN (Cat# S-2000, Vector) or 5% goat serum for ΔFosB (Cat# S-1000, Vector), diluted in a solution of 0.25% (volume/volume or v/v) Triton-X 100, and 1% (weight/volume or w/v) bovine serum albumin in 0.1 M TB. Sections were then incubated overnight at 4°C in primary antibody to NeuN (mouse monoclonal, 1:5000; Cat# MAB377, Chemicon International) or anti-ΔFosB (rabbit monoclonal, 1:1000; Cat# D3S8R, Cell Signaling), diluted in a solution of 0.25% (v/v) Triton-X 100, and 1% (w/v) bovine serum albumin in 0.1 M TB. Both NeuN and ΔFosB have been well-characterized as antigens (Mullen et al. 1992 [\[1\]](#), Wolf et al. 1996 [\[2\]](#), Chen et al. 1997 [\[3\]](#), Sarnat et al. 1998 [\[4\]](#)) and the antibodies we used have been commonly employed in the past (Chen et al. 1997 [\[3\]](#), Duffy et al. 2013 [\[5\]](#), Corbett et al. 2017 [\[6\]](#)).

On the following day, sections were washed in 0.1 M TB (3 x 5 min) and then incubated for 60 min in biotinylated horse anti-mouse secondary antibody for NeuN (1:500, Cat# BA-2000, Vector) or biotinylated goat anti-rabbit secondary antibody for ΔFosB (1:500, Cat# BA-1000), diluted in a solution of 0.25% (v/v) Triton-X 100, and 1% (w/v) bovine serum albumin in 0.1 M TB. The sections were then washed in 0.1 M TB (3 x 5 min) and incubated in avidin-biotin complex for 2 hrs (1:1000; Cat# PK-6100, Vector). They were washed in 0.1 M TB (3 x 5 min) and then reacted in a solution containing 0.5 mg/mL 3, 3'-diaminobenzidine (DAB; Cat# D5905, Sigma-Aldrich), 40 µg/mL ammonium chloride (Cat# A4514, Sigma-Aldrich), 25 mg/mL D(+)-glucose (Cat# G5767, Sigma-Aldrich), and 3 g/mL glucose oxidase (Cat# G2133, Sigma-Aldrich) in 0.1 M TB. This method slowed the reaction time so that the reaction could be temporally controlled, meaning that the reaction could be stopped when the immunoreactivity was robust but background was still low. The sections were then washed in 0.1 M TB (3 x 5 min), mounted on gelatin-coated slides (1% bovine gelatin, Cat# G9391, Sigma-Aldrich) and dried at RT overnight. The following day they were dehydrated in increasing concentrations of ethanol (90%, 10 min; 95% 10 min; 100%, 10 min; 100% again, 10 min), washed in Xylene (10 min; Cat# 534056, Sigma-Aldrich) and cover-slipped with Permount (Cat# 17986-01, Electron Microscopy Sciences).

2. Analysis

Photomicrographs were acquired using ImagePro Plus V7.0 (Media Cybernetics) and a digital camera (Model RET 2000R-F-CLR-12, Q-Imaging). NeuN and ΔFosB staining were quantified from micrographs using ImageJ (V1.44, National Institutes of Health). All images were first converted to grayscale and in each section, the hilus was traced, defined by zone 4 of Amaral (1978). A threshold was then calculated to identify the NeuN-stained cell bodies but not background. Then NeuN-stained cell bodies in the hilus were quantified manually. Note that the threshold was defined in ImageJ using the distribution of intensities in the micrograph. A threshold was then set to identify the dark cells but not the background slide using A slider was pushed from the low

level of staining corresponding to background to the location where staining intensity makes a sharp rise, reflecting strongly stained cells. Below that was considered background and above that was defined as above threshold.

To count Δ FosB-stained cells, images were converted to grayscale and in each section, the dentate gyrus GCL was outlined and defined as a region of interest (ROI). A threshold was then set to identify the dark cells but not the background, analogous to the method to define threshold in the preceding paragraph. Two thresholds were used. The first threshold was most inclusive of immunoreactive cells. Presumably this group reflected cells with less neuronal activity. Then counts were made a second time with a higher threshold. This group probably corresponded to cells with the highest activity. ImageJ was used to calculate the area (in pixels) within the ROI that was above threshold.

D. Video-electroencephalographic (video-EEG) recordings

1. Stereotaxic surgery

EEG electrodes (stainless steel jeweler's screws) were implanted in Tg2576 animals at 4 weeks of age. WT littermates were not implanted because previous work from our laboratory has shown that WT mice have no IIS (Kam et al. 2016 [\[link\]](#)). The animals were anesthetized by isoflurane inhalation (3% isoflurane, 2% oxygen for induction). For maintenance during surgery, isoflurane was $\leq 1.75\%$ and flow rate was 1.2 L/min. Mice were placed in a stereotaxic apparatus (David Kopf Instruments). Prior to the implantation of the electrodes, animals were injected with the analgesic Buprenex (0.2 mg/kg, s.c.; buprenorphine hydroxide, NDC# 12496-0757-5, Reckitt Benckiser), which was diluted in saline (0.03 mg/mL in sterile 0.9% sodium chloride solution, Vedco Inc.). The skull was exposed with a midline incision and six holes were drilled for the placement of subdural screw electrodes (Cat# 8209, 0.10" stainless steel screws, Pinnacle Technology).

The coordinates for the electrode placement were: right occipital cortex (AP - 3.5 mm, ML 2.0 mm), left frontal cortex (AP -0.5 mm, ML -1.5 mm), left hippocampus (AP -2.5 mm, ML -2.0 mm), and right hippocampus (AP -2.5 mm, ML 2.0 mm (Kam et al. 2016 [\[link\]](#)). Two additional screws were placed, one over the right olfactory bulb as ground (AP 2.3 mm, ML 1.8 mm) and another at the midline over the cerebellum as a reference (relative to Lambda: AP -1.5 mm, ML -0.5 mm). Screws were attached to an eight-pin connector (Cat# ED85100-ND, Digi-Key Corporation), which was placed over the skull and secured with dental cement (Cat# 4734FIB, Lang Dental Mfg. Co., Inc.).

After surgery, animals were placed on a heating blanket overnight and injected with lactated Ringer's solution (50 mL/kg at 31°C; NDC# 099355000476, Aspen Veterinary Resources Ltd). They were then transferred to the room where the video-EEG is recorded. Implanted animals were allowed to recover for one week following the surgery.

2. Video-EEG recording

Video-EEG was recorded starting at 1.25 months (5 weeks) of age. The animals were then recorded at 2, 3, 4, 5 and 6 months of age. Each recording session lasted 24 hr so that a long period of sleep could be acquired, since IIS occur primarily in sleep (Kam et al. 2016 [\[link\]](#)).

Mice were placed into 21 cm x 19 cm transparent cages with food and water provided *ad libitum* and corncob bedding. A pre-amplifier was inserted into the eight-pin connector on the skull, which was connected to a multichannel commutator (Cat# 8408, Pinnacle Technology). This arrangement allowed for free range of movement throughout the recording. EEG signals were acquired at 2000 Hz and bandpass filtered at 0.5 – 200 Hz using Sirenia Acquisition V2.0.4, Pinnacle Technology). Simultaneous video recordings were captured using an infrared camera (Cat# AP-DCS100W, Apex CCTV).

3. Analysis

EEG recordings were analyzed offline with Neuroscore V3.2.1 (Data Science International). IIS were defined as large amplitude, 10-75 msec deflections occurring synchronously in all four leads (Kam et al. 2016 [DOI](#)). They were quantified by first setting an amplitude threshold (calculated separately for each recording) and a duration criterion (10-75 msec). To determine the amplitude threshold, we calculated the root mean square (RMS) amplitude for noise in each recording during a 60 sec segment of baseline activity that did not include any artifacts. The threshold for IIS was set at 9 standard deviations above the RMS amplitude. This threshold was selected because it identified IIS extremely well and excluded movement or other artifacts. The IIS in one of the hippocampal channels was used because of data suggesting IIS begin in the hippocampus (Kam et al. 2016 [DOI](#)). Following the automatic analysis of the IIS, the software's detection accuracy was verified manually for each recording to ensure that all spikes appeared in all four channels and that no artifacts were included in the final IIS count.

E. Statistical comparisons

Data are expressed as mean $\pm$ standard error of the mean (SEM). P-value was set to <0.05 prior to all experiments. Parametric data comparing two groups used unpaired two-tailed t-tests. For >2 groups, one-way ANOVA was used followed by Tukey-Kramer post-hoc tests that corrected for multiple comparisons. For data with two main factors, two-way ANOVA was followed by Tukey-Kramer post-hoc tests. Interactions are not reported unless they were significant.

For the analysis of IIS frequency, a repeated measures ANOVA was not possible because animal numbers were not the same at all ages. This was due to mortality in low choline-treated mice or other unidentified reasons. Therefore, a mixed effects analysis was conducted.

For non-parametric data, Mann-Whitney *U* tests were used to compare two groups and Kruskal-Wallis for >2 groups. The post-hoc tests were Dunn's test for multiple comparisons. Tests were conducted using Prism (V 9.4.1, Graphpad Software).

III. RESULTS

A. Behavior

NOL and the NOR, unlike other tasks previously used to evaluate cognitive deficits in >9 months-old Tg2576 mouse model (i.e., Morris Water Maze, Radial Arm Water Maze, Y-Maze; Yassine et al. 2013 [DOI](#), Wolf et al. 2016 [DOI](#)), provided an opportunity to capture cognitive deficits in animals as young as three months-old (Duffy et al. 2015 [DOI](#)). **Figure 1B-D** [DOI](#) shows a schematic of the specific experimental procedures for NOL and NOR.

1. NOL

The results of the NOL task are presented in **Figure 2** [DOI](#). Exploration of the novel object is expressed as a percentage of the total exploration of both objects. During the training session, animals did not show preference for one object over the other and this result was independent of genotype. Thus, a two-way ANOVA revealed no main effects of genotype on novel object exploration during training at three ($F(1,97) = 0.40$, $p = 0.53$) or six months of age ($F(1,70) = 1.94$, $p = 0.17$). There also was no effect of maternal diet on training at three ($F(2,97) = 0.06$, $p = 0.94$) or six months of age ($F(2,70) = 2.94$, $p = 0.06$).

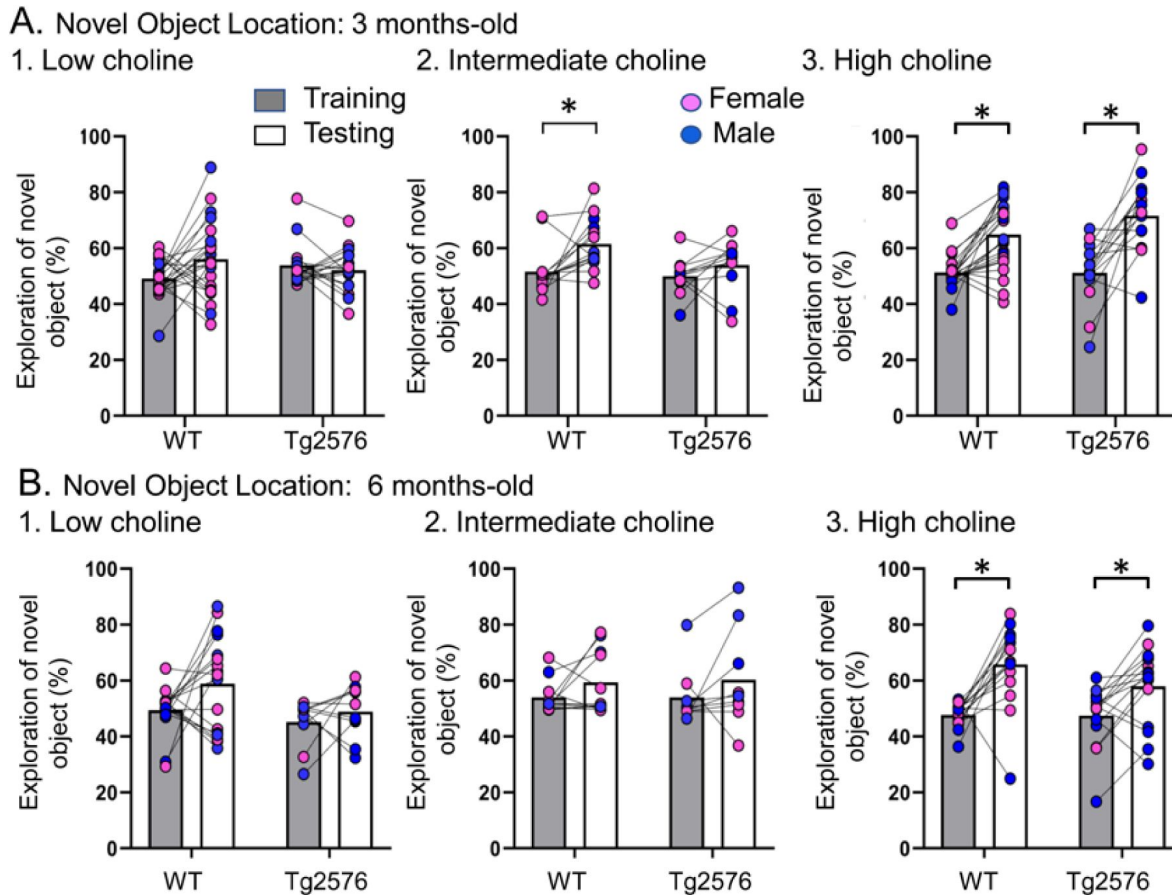


Figure 2.

Choline enrichment reduced spatial memory deficits.

A. Three months-old mice.

1. WT and Tg2576 mice exposed to the low choline diet showed spatial memory deficits.
2. Tg2576 exposed to the intermediate diet showed spatial memory deficits but not WT.
3. Spatial memory was improved in animals exposed to the high choline diet.

B. Six months-old mice.

1. WT and Tg2576 mice exposed to the low choline diet showed spatial memory deficits.
2. WT and Tg2576 mice exposed to the intermediate diet showed spatial memory deficits.
3. WT and Tg2576 mice exposed to the high choline diet had improved spatial memory.

Next, we asked whether Tg2576 mice were deficient in the test phase of NOL. First we studied the 3 months-old mice. We used a two-way ANOVA with genotype and task phase (training versus testing) as main factors. In animals that received the intermediate diet, a two-way ANOVA revealed a significant main effect of the phase of the task (training or testing; $F(1,24) = 7.48$, $p = 0.01$). Consistent with previous studies (Duffy et al. 2015 [↗](#)), post-hoc analyses revealed a significant increase in the exploration of the novel object during testing in WT ($p = 0.02$) but not Tg2576 mice ($p = 0.50$; **Figure 2A2** [↗](#)). Thus, Tg2576 mice were impaired in NOL when fed the intermediate diet.

In animals that received the high choline diet, a two-way ANOVA revealed a main effect of the task phase ($F(1,36) = 48.23$, $p < 0.001$). Post-hoc analyses revealed a significant increase in the exploration of the novel object during testing in both WT ($p < 0.001$) and Tg2576 mice ($p < 0.001$; **Figure 2A3** [↗](#)). Therefore, high choline supplementation improved memory in Tg2576 mice.

In animals treated with the low choline diet, there was no effect of phase of the task ($F(1,37) = 1.39$, $p = 0.25$). These data suggesting that relatively low levels of choline during early life impaired spatial memory in both WT and Tg2576 mice (**Figure 2A** [↗](#)). Thus, low choline had a significant adverse effect. Adverse effects are further supported by survival plots showing that there was more mortality at earlier ages in offspring exposed to the low choline diet (**Supplemental Figure 1** [↗](#)).

Figure 2B [↗](#) shows the results in 6 months-old WT and Tg2576 mice. In animals that received the intermediate diet, a two-way ANOVA showed no main effect of genotype ($F(1,18) = 0.01$, $p = 0.92$) or task phase ($F(1,18) = 3.40$, $p = 0.08$), revealing age-related memory deficits even in WT mice at six months of age (**Figure 2B** [↗](#)). In contrast, the high choline group showed a main effect of task phase ($F(1,29) = 28.32$, $p < 0.001$). Post-hoc analyses showed a significant increase in the exploration of the novel object during testing in both WT ($p < 0.001$) and Tg2576 ($p = 0.02$; **Figure 2B** [↗](#)). In animals that received the relatively low choline diet, there was no effect of task phase ($F(1,25) = 3.00$, $p = 0.10$; **Figure 2B** [↗](#)). Thus, low choline had adverse effects on NOL in both WT and Tg2576 mice.

Taken together, these results demonstrate the lasting beneficial effects of the high choline diet and adverse effects of low choline on offspring in a spatial memory task in Tg2576 as well as WT littermates.

2. NOR

We first confirmed that animals did not show preference for one object over the other in training. A two-way ANOVA revealed no main effects of genotype at three months ($F(1,71) = 2.59$, $p = 0.54$) and no effect of diet ($F(2,71) = 0.22$, $p = 0.81$). The same result was obtained at six months of age (genotype: ($F(1,66) = 0.11$, $p = 0.75$); diet: ($F(2,66) = 0.98$, $p = 0.38$). Thus, novel object exploration during training approached 50% in all treatment groups, independent of genotype and maternal diet, and at both ages.

Figure 3A [↗](#) shows the results of the testing phase of the NOR task in 3 months-old WT and Tg2576 mice. In animals that received the intermediate choline diet, a two-way ANOVA revealed a significant main effect of task phase ($F(1,19) = 40.53$, $p < 0.01$) and no effect of genotype ($F(1,19) = 0.78$, $p = 0.39$). Post-hoc analyses revealed a significant increase in the exploration of the novel object during testing compared to training for both WT ($p < 0.01$) and Tg2576 ($p < 0.01$; **Figure 3A** [↗](#)). These results suggest that Tg2576 did not have a deficit in NOR at 3 months of age.

Mice fed the high choline diet also showed no deficit in NOR at 3 months of age. Thus, a two-way ANOVA revealed a main effect of task phase ($F(1,26) = 17.51$, $p < 0.01$) but not genotype ($F(1,26) = 3.53$, $p = 0.07$). Post-hoc analyses revealed a significant increase in the exploration of the novel object during testing in both WT ($p < 0.01$) and Tg2576 ($p = 0.03$; **Figure 3A** [↗](#)).

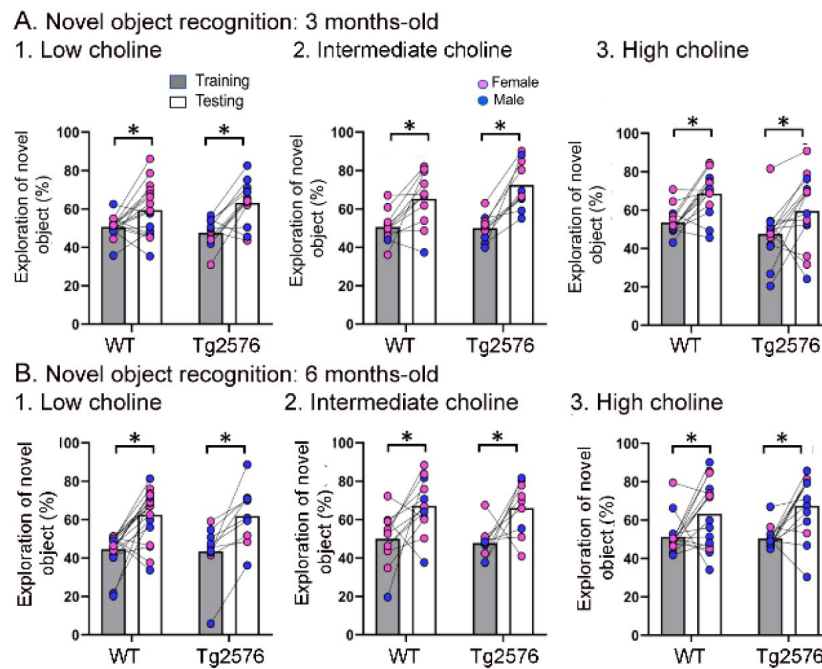


Figure 3.

WT and Tg2576 mice showed recognition memory regardless of diet.

A. Three months-old mice.

1-3. WT and Tg2576 mice treated performed the NOR task.

B. Six months-old mice.

1-3. WT and Tg2576 mice performed the NOR task.

In animals treated with the low choline diet in early life, there was no impairment in NOR at 3 months of age (task phase ($F(1,26) = 22.61$, $p < 0.01$; genotype ($F(1,26) = 0.02$, $p = 0.89$). There was a significant increase in the exploration of the novel object in testing both in WT ($p = 0.03$) and Tg2576 ($p < 0.01$).

Figure 3B shows that there were no impairments at 6 months of age. In animals that received the intermediate diet, a two-way ANOVA revealed a significant main effect of task phase ($F(1,19) = 13.65$, $p < 0.01$) but no main effect of genotype ($F(1,19) = 0.28$, $p = 0.60$). Animals explored the novel object significantly more during testing in WT ($p = 0.03$) and Tg2576 mice ($p = 0.04$; **Figure 3B**). In animals that received the other diets, results were the same. For the high choline group, there was an effect of task phase ($F(1,27) = 16.26$, $p < 0.01$) but not genotype ($F(1,27) = 0.24$, $p = 0.63$) and a significant increase in the exploration of the novel object during testing in both WT ($p = 0.04$) and Tg2576 ($p < 0.01$; **Figure 3B**). Similarly, in animals treated with the low choline diet, there was a main effect of task phase ($F(1,26) = 32.66$, $p < 0.01$), no effect of genotype ($F(1, 6) = 0.05$, $p = 0.82$), and more exploration of the novel object during the test phase (WT, $p < 0.01$; Tg2576, $p < 0.01$).

Thus, at both 3 and 6 months of age, WT and Tg2576 mice performed well in the NOR task. These results suggest that young Tg2576 mice are less sensitive to NOR than NOL. The greater sensitivity to NOL is consistent with past demonstrations that the dentate gyrus contributes to NOL (Sahay et al. 2011, Kesner et al. 2015, Spyra & Hess 2018, Vandrey et al. 2020, Gulmez Karaca et al. 2021, GoodSmith et al. 2022), and early signs of adverse effects in the dentate gyrus in mouse models of AD (impaired LTP; Jacobsen et al. 2006, altered gene expression; Palop et al. 2011). Our implementation of NOL may have increased the dentate gyrus-dependence of the task by making the object locations relatively close together, because the dentate gyrus is critical to distinguishing small differences in locations. In studies by Pofahl and colleagues, it was shown that distances between objects like those we used made NOL dentate gyrus-dependent (Pofahl et al. 2021).

3. Exploration time

Total object exploration (TOE) was measured to address any effects of genotype or diet on the total duration of exploration of objects (**Figure 4**).

a. 3 months of age

For NOL at 3 months of age, there were effects of genotype ($F(1,67) = 6.891$, $p = 0.01$) but not maternal diet ($F(2,87) = 0.67$, $p = 0.63$; **Figure 4A**). Tukey-Kramer post-hoc tests showed that Tg2576 mice fed the low choline diet showed more exploration than WT mice fed the same diet but the effect was on the border of significance ($p = 0.049$; data not shown). Three months-old mice tested with NOR showed no significant effect of genotype ($F(1,76) = 1.35$, $p = 0.25$) or diet ($F(2,76) = 0.30$, $p = 0.61$; **Figure 4C**).

b. 6 months of age

1) Exploration

NOL and NOR showed significant effects of diet on exploration at 6 months of age (**Figure 4B,D**). For NOL, there was no effect of genotype ($F(1,66) = 0.33$; $p = 0.57$) but there was an effect of maternal diet ($F(2,66) = 8.82$; $p = 0.0004$). For NOR, the same was true (genotype: $F(1,72) = 0.96$, $p = 0.33$; diet: $F(2,72) = 8.50$, $p = 0.0005$). We then pooled genotypes to determine which diets differed. For NOL, mice treated with the relatively low choline diet had less exploration than mice fed the high choline diet (one-way ANOVA, $F(2,77) = 6.90$; $p = 0.002$; Tukey post-hoc test, $p = 0.005$; **Figure 4B**) and the mice that were fed the intermediate diet also had less exploration than mice fed the high choline diet (Tukey post-hoc test, $p = 0.008$; **Figure 4B**). Results for NOR were the same:

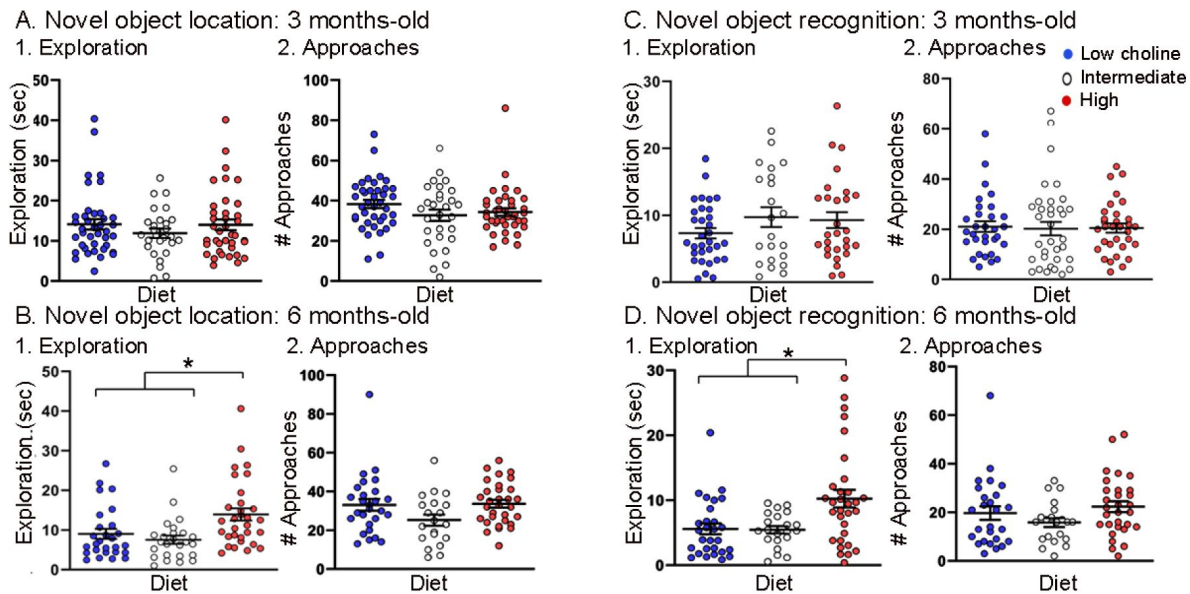


Figure 4.

Tg2576 mice treated with the choline supplemented diet spent more time with the objects at each approach.

A-B. NOL.

There was no effect of diet on time exploring or approaches to the novel object in the NOL task at 3 months of age (A) but there were significant differences at 6 months of age (B). Mice exposed to the high choline diet spent more time exploring than mice that had been fed the low choline or the intermediate diet.

C-D. NOR.

There was no effect of diet on time exploring or approaches to the novel object in the NOR task at 3 months of age (C) but there were significant differences at 6 months of age (D). Mice that had been fed the high choline diet spent more time exploring than mice that had been fed the low choline or the intermediate diet.

mice treated with the low choline diet had less exploration than mice fed the high choline diet (one-way ANOVA, $F(2,74) = 4.81$; $p = 0.02$; Tukey post-hoc test, $p = 0.02$; **Figure 4D**) and the mice that were fed the intermediate diet also had less exploration than mice fed the high choline diet (Tukey post-hoc test, $p = 0.03$; **Figure 4D**).

2) Approaches

To gain insight into the potential reason for differences in exploration, we measured total object approaches (TOA; **Figure 4A-D**). TOA was defined as the number of approaches to the familiar object + number of approaches to the novel object. There were no differences by genotype (two-way ANOVA, $F(1,73) = 2.930$, $p = 0.09$) or maternal diet ($F(2,73) = 2.36$, $p = 0.10$; **Figure 4A-D**).

Taken together, the results indicate that, in 6 months-old mice, animals that received the high choline diet spent more time with objects at each approach. This may underlie increased object location memory in high choline-treated mice by increasing time for information processing during an approach. It may also help in the NOR task with object recognition memory, although it did not seem impaired in mice fed other diets. Another possibility is that as Tg2576 mice age they show compensatory changes that enhance memory. This has been suggested for procedural learning (Middei et al. 2004) but not for memory of objects. A third possibility is that the high choline diet reduces anxiety (Glenn et al. 2012, Langley et al. 2015, McCall et al. 2015). Reduced anxiety may lessen fear of exploring objects, and as a result, animals may spend more time with objects.

For 3 months-old mice there were no effects of maternal diet on exploration, suggesting that increased time with objects for a given approach is a change that develops with age. To our knowledge, this age-related change has not been previously reported. It could be a strategy to cope with impairments that are developing as AD pathology manifests.

B Anatomy

1. NeuN

To further analyze the effects of dietary choline early in life, we used an antibody against a neuronal nuclear antigen, NeuN. Previous studies have found that reduced expression of NeuN often occurs in neurons after insults or injury (e.g., ischemia, toxicity and even aging; Lind et al. 2005, Portiansky et al. 2006, Buckingham et al. 2008, Kadriu et al. 2009, Matsuda et al. 2009, Won et al. 2009, Duffy et al. 2011). For example, when NeuN is reduced after a reduction in ankyrin-rich membrane spanning kinase D-interacting substrate of 220K, ARMS/Kidins, a protein critical to neurotrophin signaling, neurons in the entorhinal cortex demonstrate signs of toxicity such as pkyrosis (Duffy et al. 2011, Duffy et al. 2015). Since NeuN is reduced when A β levels are elevated (Wu et al. 2016) and NeuN is reduced in AD patients (Camporez et al. 2021), we used NeuN expression to ask if the high choline diet could rescue the loss of NeuN-ir.

As shown in **Figure 5**, relatively weak NeuN-ir was observed in Tg2576 animals in the hilus, an area known to be highly vulnerable to insult and injury (Scharfman 1999). The most NeuN loss appeared to be in mice that received the low choline diet and the intermediate diet compared to the high choline diet (**Figure 5B**). Therefore, we conducted a two-way ANOVA with diet and septotemporal levels as main factors. To examine the septotemporal axis, quantification from 2-3 dorsal coronal sections were averaged to provide a value for dorsal DG and 2-3 ventral horizontal sections were averaged for a ventral value. There was no effect of diet ($F(2,30) = 2.11$, $p = 0.14$) nor septotemporal level ($F(1,30) = 0.02$, $p = 0.88$) but there was a significant interaction of diet and septotemporal level ($F(2,30) = 3.88$, $p = 0.04$). A one-way ANOVA using dorsal values showed a significant effect of diet ($F(2,21) = 7.58$, $p = 0.003$) but this was not the case for ventral values

($F(2,23) = 0.13$; $p = 0.88$; **Figure 5D**). In the dorsal DG, there was less NeuN-ir in mice fed the low choline diet versus the high choline diet (Tukey-Kramer post hoc test, $p = 0.03$) and less NeuN in mice fed the intermediate vs high choline diet ($p = 0.003$; **Figure 5D**). The results suggest that choline enrichment protected hilar neurons from NeuN loss in Tg2576 mice.

2. Δ FosB

To complement the information from the video-EEG recordings we used a marker of elevated neuronal activity, Δ FosB (**Figure 6**). Δ FosB is a truncated variant of the transcription factor FosB, which is increased by enhanced neuronal activity; Δ FosB has a half-life of approximately 8 days (Ulery-Reynolds et al., 2009), so when Δ FosB is elevated, it reflects increased neuronal activity over the last 10-14 days (McClung et al. 2004). Previous studies have shown that when the J20 mouse model of AD is examined with an antibody to Δ FosB, the GC layer shows extremely high levels of Δ FosB expression (Corbett et al. 2017), similar to a mouse with chronic spontaneous seizures (epilepsy; Morris et al. 2000). This is not surprising since the J20 mice have recurrent seizures (Palop et al. 2007). Therefore, we asked if Tg2576 mice would have robust Δ FosB in the GC layer, and choline supplementation would reduce it.

There was strong expression of Δ FosB in Tg2576 GCs especially in mice fed the low choline diet (**Figure 6**). Two-way ANOVA showed a main effect of maternal diet ($F(2,18) = 3.76$, $p = 0.04$) but there was not a main effect of septotemporal level ($F(1,18) = 1.3$, $p = 0.26$).

To clarify the effect of maternal diet, a one-way ANOVA was conducted. There was an effect of diet ($F(2,9) = 10.74$, $p = 0.004$) with post-hoc tests showing significantly less GC Δ FosB expression in high choline-treated mice compared to low choline ($p = 0.003$). Low choline-treated mice were not significantly different from mice that were fed the intermediate diet ($p = 0.08$). These data used dorsal sections so we repeated the analysis for the ventral DG, and there also was no significant effect of diet ($F(2,9) = 0.79$, $p = 0.52$; **Figure 6**).

Δ FosB quantification was repeated with a lower threshold to define Δ FosB-ir GCs (see Methods) and results were the same (**Figure 6**). One-way ANOVA showed a significant effect of diet ($F(2,41) = 14.66$, $p < 0.0001$), and low choline mice had greater Δ FosB-ir than high choline mice ($p = 0.003$). Mice fed the intermediate diet had higher expression than mice fed the high choline diet ($p = 0.0001$). These data suggest that high choline in the diet early in life can reduce hyperexcitability of GCs in offspring later in life. In addition, low choline has an opposite effect, again suggesting this maternal diet has adverse effects.

C. IIS

Previous research has shown that Tg2576 mice exhibit IIS (**Figure 7A**) starting at very young ages: 4 weeks (Kam et al. 2016) or 6 weeks (Bezzina et al. 2015). Therefore, we performed video-EEG using cortical (left frontal cortex, right occipital cortex) and hippocampal (left and right) electrodes in WT and Tg2576 mice (**Figure 7B-C**). Animals were recorded for 24 hr each session so that there was sufficient time to capture major behavioral states (exploration, awake rest, and sleep). Consistent with previous studies (Bezzina et al. 2015, Kam et al. 2016) we observed IIS in Tg2576 mice but not WT littermates.

As shown in **Figure 7**, animals that received the intermediate diet had a significantly higher number of IIS in the 24 hr recording periods compared to animals that received the high choline and the low choline diets. A mixed model analysis showed that there was a significant effect of age ($F(2,37) = 3.38$; $p = 0.04$) and maternal diet ($F(2,36) = 8.12$; $p = 0.01$). At the 1.25-month recording, Tukey post-hoc analyses showed that IIS frequency in animals treated with the intermediate diet was higher compared to animals treated with the low ($p = 0.03$) or high choline diets ($p = 0.04$). This also was true for 2 months and 3 months (low choline, high choline, $p < 0.05$). At 4 months, the

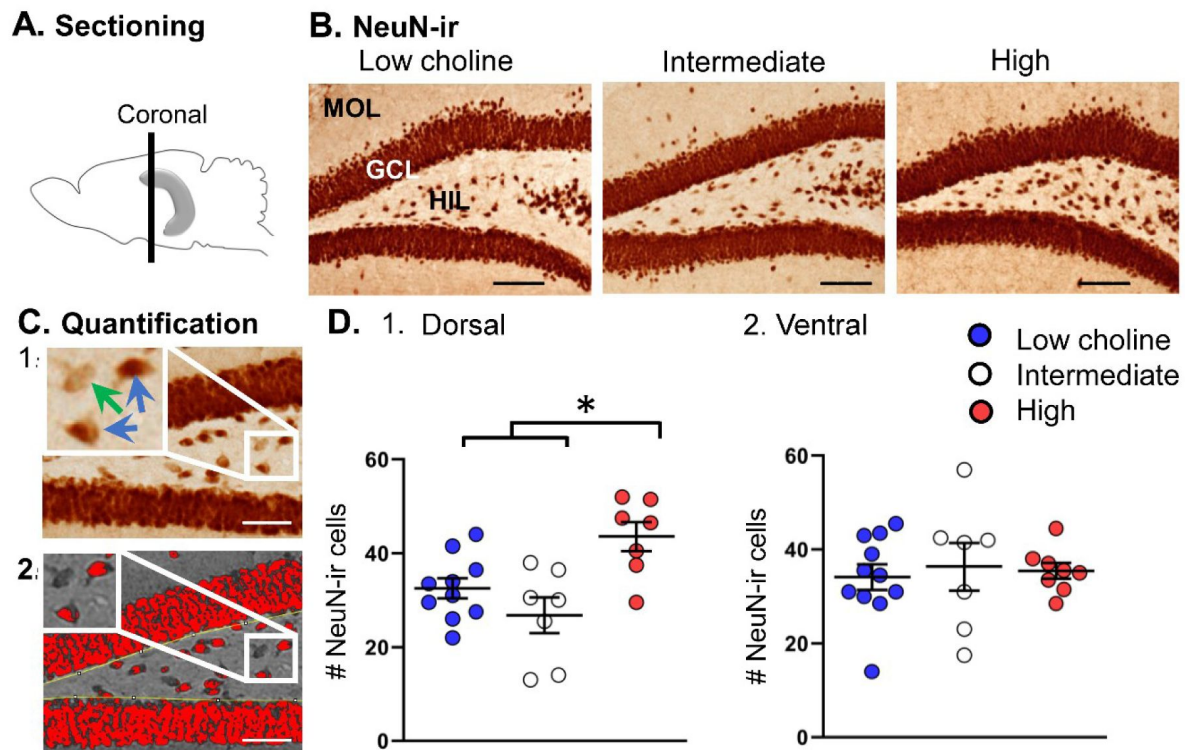


Figure 5.

Choline supplementation improves NeuN immunoreactivity (ir) in hilar cells in Tg2576 animals.

A. A sagittal view of the brain illustrating the coronal sectioning plane.

B. Representative images of NeuN-ir staining in the dorsal dentate gyrus of Tg2576 animals from each treatment group.

C. Quantification methods. Representative images demonstrating the thresholding criteria established to quantify NeuN-ir.

1. A NeuN-stained section shows an inset (arrow) expanded at the top left to show 3 hilar cells. NeuN-ir cells are marked by blue arrows. NeuN-ir that was too close to background staining to be considered positive is marked by a green arrow.

2. After converting the image to grayscale, a threshold was set so that all cells stained well with NeuN antibody were red. The inset shows that the two cells that were darkest met the threshold while the weakly stained cell did not.

D. Six months-old Tg2576 mice treated with the high choline diet had significantly more NeuN-ir cells in the hilus in the dorsal dentate gyrus (1) compared to other treatment groups but not ventral dentate gyrus (2).

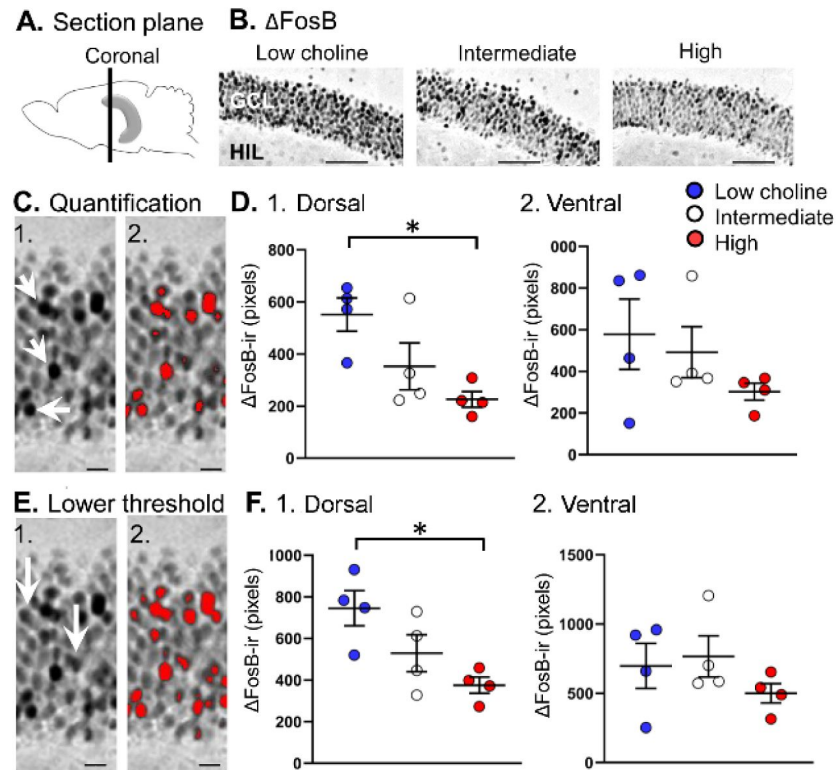


Figure 6.

Choline supplementation reduced Δ FosB expression in dorsal GCs of Tg2576 mice.

- A.** A sagittal view of the brain illustrating the coronal sectioning plane.
- B.** Representative images of Δ FosB staining in the anterior dentate gyrus of Tg2576 animals from each treatment group.
- C.** Quantification methods. Representative images demonstrating the thresholding criteria established to quantify Δ FosB.
1. A Δ FosB-stained section shows strongly-stained cells (white arrows).
2. A strict thresholding criteria was used to make only the darkest stained cells red.
- D.** Six months-old Tg2576 mice treated with the choline supplemented diet had significantly more Δ FosB-ir pixels above threshold in the dorsal dentate gyrus (1) compared to other treatment groups but not ventral dentate gyrus (2).
- E.** Methods are shown using a threshold that was less strict.
1. Some of the stained cells that were included are not as dark as those used for the strict threshold (white arrows).
2. All cells above the less conservative threshold are shown in red.
- F.** With the less conservative threshold, the results were the same as D. Mice that were fed the high choline diet had less Δ FosB-ir pixels than the mice that were fed the other diets. However, this was true for dorsal (1) but not ventral (2) sections.

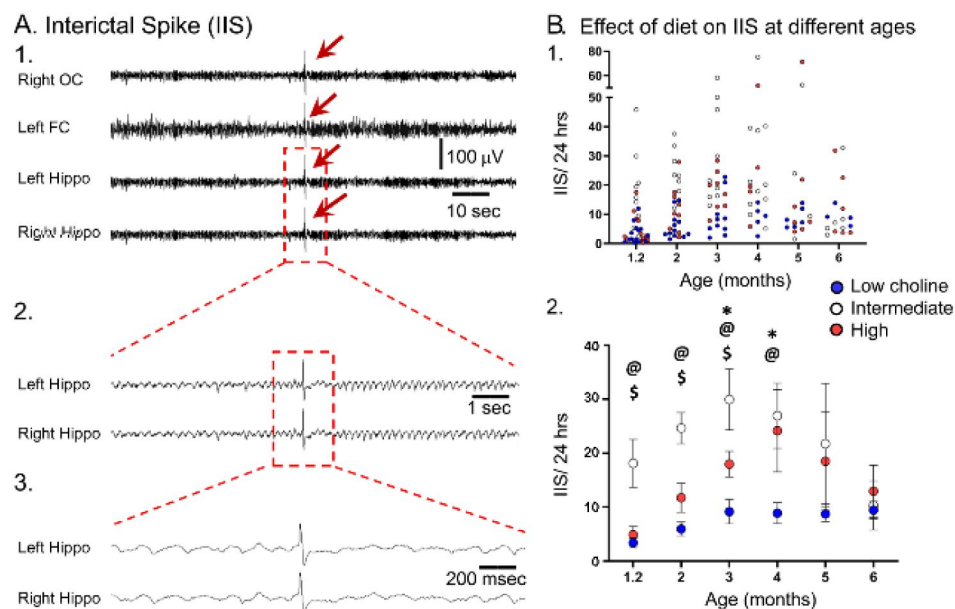


Figure 7.

The high choline diet reduced IIS frequency in Tg2576 animals.

A. 1. Representative example of an IIS (red arrow). IIS were defined as occurring in all four channels (generalized) to distinguish them from focal spikes that can often be artifact. 2-3. The IIS shown in A1 is expanded.

B. 1. Scatter plot of IIS frequency at each age of recording.

1. Means and sem are plotted. The high choline diet group had fewer IIS than the intermediate diet group at ages 1-3 months (\$, $p < 0.05$) and the low choline group had less IIS than the intermediate diet at ages 1-4 months (@, $p < 0.05$). The high and low choline diet groups were significantly different at ages 3 and 4 months (*, $p < 0.05$).

low choline groups had significantly reduced IIS frequency compared to the intermediate diet ($p = 0.009$) but not the high choline group ($p = 0.98$). At 5-6 months, IIS frequencies were not significantly different in the mice fed the maternal diets (all $p > 0.05$).

As shown previously (Kam et al. 2016 [↗](#)), and in **Figure 7** [↗](#), at 4-6 months of age there was high variability in IIS frequency between animals. It is possible that IIS frequency becomes more variable because we found that seizures occur in the Tg2576 mice after 2 months of age (Kam et al. 2016 [↗](#)), and there was a sharp increase in IIS during the day before and after a seizure (**Supplemental Figure 3** [↗](#)).

D. Seizures and premature mortality in mice fed the low choline diet

We found mice fed the low choline diet had greater Δ FosB-ir in GCs and the hilus showed very low NeuN-ir. Therefore we asked whether low choline-treated mice had more seizures than mice fed the other diets. We recorded 8 mice by video-EEG for 5 days each (at 6 months of age) to examine seizures and found 2 mice from the low choline group had seizures (11 seizures over 2 of the 5 days in one mouse, 1 seizure in the other mouse) whereas none of the other mice had seizures ($n=0/4$, 2 intermediate and 2 high choline, data not shown).

These values are probably an underestimate for the low choline group because many mice in this group appeared to die in a severe seizure prior to 6 months of age (**Supplemental Figure 2** [↗](#)). Therefore, the survivors at 6 months probably were the subset with few seizures.

The reason that low choline-treated mice appeared to die in a seizure was that they were found in a specific posture in their cage which occurs when a severe seizure leads to death (**Supplemental Figure 2** [↗](#)). They were found in a prone posture with extended, rigid limbs (**Supplemental Figure 2** [↗](#)). Regardless of how the mice died, there was greater mortality in the low choline group compared to mice that had been fed the high choline diet (**Supplemental Figure 2** [↗](#)).

E. Sex differences

As shown in **Figures 2** [↗](#)-**3** [↗](#), there appeared to be no sex differences in NOL or NOR. For confirmation, we studied NOL further, since that was the task that showed effects of diet. A three-way ANOVA with sex, genotype and task phase as factors showed no significant effect of sex at three months of age (low choline diet, $F(1,35) = 0.0001$; $p = 0.99$). Results were the same for the three months-old mice fed the intermediate diet ($F(1,22) = 0.062$; $p = 0.81$) and three months-old mice that were fed the high choline diet ($F(1,34) = 0.43$; $p = 0.52$). At six months of age there also were no significant effects of sex (low choline diet, $F(1,23) = 0.34$; $p = 0.57$; intermediate diet, $F(1,16) = 2.49$; $p = 0.13$; high choline diet, $F(1,27) = 0.29$; $p = 0.87$).

Regarding IIS, we did not find sex differences (**Supplemental Figure 4** [↗](#)). A two-way ANOVA with sex and diet as factors showed no effect of sex ($F(1,20) = 2.48$; $p = 0.13$) or diet ($F(1,20) = 3.56$; $p = 0.07$). Note that for this comparison we used the low choline and high choline groups only because sex information was only available for these mice. The results for months 2-3 also showed no effect of sex (2 months: $F(1,17) = 0.64$; $p = 0.43$; 3 months: $F(1,15) = 0.19$; $p = 0.67$).

IV. DISCUSSION

Summary

This study showed that early life choline supplementation had several beneficial effects in Tg2576 mice. The high choline diet led to reduced IIS frequency, improved behavior in the NOL task, and improved expression of Δ FosB.

We showed for the first time that Tg2576 mice have pathology in the hilus at six months of age in that hilar neurons are deficient in immunostaining with an anti-NeuN antibody. Tg2576 offspring on the high choline diet showed a restoration of hilar NeuN staining. Because hilar neurons play a role in NOL (Bui et al. 2018 [\[1\]](#)), it is possible that rescue of hilar neurons by the high choline diet led to improved spatial memory. Hilar neurons have been suggested to contribute to excitability of GCs (Sloviter 1994 [\[2\]](#), Scharfman 1999 [\[3\]](#), Scharfman & Myers 2012 [\[4\]](#), Jinde et al. 2013 [\[5\]](#), Bui et al. 2018 [\[1\]](#), Botterill et al. 2019 [\[6\]](#)), so phenotypic rescue of hilar neurons may also have contributed to the ability of the high choline diet to reduce hyperexcitability.

We also found surprising effects of treating mice with relatively low choline in the maternal diet. On the one hand, IIS frequency was reduced. However, these mice showed premature mortality. Furthermore, in WT mice where no impairments in NOL normally occur, WT mice fed the low choline diet were impaired. In Tg2576 mice, the group that had the low maternal choline diet had the highest Δ FosB expression and the lowest NeuN expression of any experimental group. Potential reasons for these surprising effects are discussed below.

Benefits of high choline

The results of this study are consistent with previous reports that prenatal or postnatal choline supplementation improves object memory. One study of iron deficiency showed that choline supplementation improved NOR (Kennedy et al. 2014 [\[7\]](#)). Wistar rats treated with a diet high in choline had improved NOR relative to rats that had been fed a diet low in choline (Moreno et al. 2013 [\[8\]](#)). In rats that were aged to 24 months, animals that had been fed a high choline diet showed improved NOR, but only in females (Glenn et al. 2008 [\[9\]](#)).

Although few studies have examined the effects of MCS on hyperexcitability, our results showing the benefits of high choline are consistent with reports that a methyl-enriched diet (including choline but also betaine, folic acid, vitamin B12, L-methionine and zinc) reduced spike wave discharges in the offspring (Sarkisova et al. 2023 [\[10\]](#)) and audiogenic seizure severity (Poletaeva et al. 2014 [\[11\]](#)). Perinatal ethanol treatment that increased excitability was mitigated by choline chloride injection from postnatal day 10-30 (Grafe et al. 2022 [\[12\]](#)).

To our knowledge our study is the first to show that MCS can exert effects on IIS and Δ FosB. However, prior studies have shown that MCS in rats improved memory after severe seizures. Using a convulsant to induce several hours of severe seizures (status epilepticus, SE), there is usually extensive hippocampal neuronal loss and when tested in the subsequent days and weeks, memory impairment. MCS reduced this impairment following SE in the Morris water maze (Yang et al. 2000 [\[13\]](#), Holmes et al. 2002 [\[14\]](#), Wong-Goodrich et al. 2011 [\[15\]](#)).

Our results are consistent with previous studies of mouse models of AD. In the APP/PS1 mouse model, it was shown that life-long choline supplementation, to the level we used (5.0 g/kg) improved memory in aged mice compared to the 1.1 g/kg diet (Velazquez et al. 2019 [\[16\]](#)). Postnatal choline supplementation also improved memory in APP/PS1 mice (Wang et al. 2019 [\[17\]](#)). Other improvements were also shown in the APP/PS1 mice, such as improved cholinergic expression via upregulation of choline acetyltransferase (Mellott et al. 2017 [\[18\]](#), Velazquez et al. 2019 [\[16\]](#)).

Adverse effects of low choline

The effects of the relatively low choline diet were surprising because several adverse effects were found. This is unexpected because 1.1 g/kg choline is not considered very low in several studies of the past, and some publications consider it normal (e.g., Lamoureux et al. 2008 [\[19\]](#)). The majority of past studies used rats (Meck et al. 1988 [\[20\]](#), Meck & Williams 1999 [\[21\]](#), 2003 [\[22\]](#), Meck et al. 2007 [\[23\]](#), Lamoureux et al. 2008 [\[19\]](#)), however, so there could be a species difference. Also, some of the past

studies used the diet for a shorter timespan than we did (e.g., [Lamoureux et al. 2008](#)) because of evidence that perinatal choline supplementation exerts its maximal effects during critical periods ([Meck & Williams 2003](#), [Meck et al. 2007](#)).

It was also surprising to find high mortality in mice exposed to the low choline diet in early life. Also, the offspring of mothers fed the low choline diet had very high Δ FosB expression in the GCs, suggesting they had chronic seizures. The observation that mice died from seizures is consistent with the Δ FosB data. There also was weak NeuN in the hilus. When WT mice were tested, NOL was impaired in the low choline group. These data suggest that the offspring of mothers fed the low choline diet were unhealthy, possibly leading to or because of seizures.

On the other hand, IIS were reduced in frequency in the same mice. There are several possible explanations for this apparent paradox. The most parsimonious explanation is the mice were unhealthy. If there was disruption of sleep, IIS might be reduced since IIS occur in sleep primarily. However, there is no evidence that MCS influences sleep in humans or laboratory animals. The one study that looked at the sleep/wake cycle in rats did not find effects ([Cheng et al. 2008](#)).

A more radical explanation is that the intermediate diet promoted IIS rather than low choline reducing IIS. The idea here is that choline is not critical to IIS because both low and high choline groups had low IIS frequency compared to the intermediate group, at least for 3 months of life. Instead of choline, a constituent of the intermediate diet may have promoted IIS. The reason why the effect of diet waned over time could have been due to the fact that after weaning all animals had the intermediate diet.

More seizures could explain premature mortality in the low choline group. However, why it would decrease IIS is not clear. One reason seizures could cause IIS to be reduced is that seizures in the Tg2576 mice typically started during sleep and woke mice from sleep. The arousal from sleep would interrupt the behavioral state when IIS usually occur as mentioned above. Also, seizures in the Tg2576 mice were followed by a depression of the EEG that would reduce IIS (postictal depression; **Supplemental Figure 3**).

NeuN

As mentioned above, NeuN is a neuronal nuclear antigen that can be phosphorylated, and when that occurs the antibody to NeuN no longer binds to NeuN. NeuN is phosphorylated in response to oxidative damage, brain injury, and toxicity ([Lind et al. 2005](#)). Therefore, it was of interest when we saw that hilar neurons of the dentate gyrus showed reduced NeuN in Tg2576 mice. Hilar neurons are mainly glutamatergic mossy cells and somatostatin (SOM)/neuropeptide Y (NPY)-expressing GABAergic neurons (HIPP cells; ([Houser 2007](#), [Scharfman & Myers 2012](#), [Scharfman 2016](#)), and both neuronal types are implicated in spatial memory functions of the GCs, as well as their excitability ([Myers & Scharfman 2009](#), [2011](#), [Scharfman & Myers 2012](#), [Jinde et al. 2013](#), [Scharfman 2016](#), [Raza et al. 2017](#), [Bui et al. 2018](#), [GoodSmith et al. 2019](#), [Li et al. 2021](#)). If damaged by the high intracellular APP/A β levels in young Tg2576 mice, the hilar neurons would be expected to show NeuN loss. This idea is consistent with deficits in SOM and NPY-stained cells ([Chan-Palay 1987](#)), indicating a vulnerability.

The reason for rescue of dorsal hilar but not ventral hilar NeuN with the high choline diet is unclear. However, there was a trend for the high choline mice to have greater NeuN in ventral tissue sections. The greater sensitivity of dorsal neurons may be related to differences in dorsal and ventral dentate gyrus gene expression patterns, since some of the genes that are differentially expressed could affect vulnerability to insult or injury ([Cembrowski et al. 2016](#), [Zhang et al. 2018](#)). Resilience of vulnerable neuronal populations, especially in the hippocampus, is a topic of high interest in AD research.

ΔFosB

Given that seizures are associated with elevated GC expression of ΔFosB (Chen et al. 1997 [↗](#), McClung et al. 2004 [↗](#), Corbett et al. 2017 [↗](#), You et al. 2017 [↗](#)), the differences in ΔFosB expression levels observed here could be the result of increased seizures in animals that received the low choline diet. Therefore, we suggest that the high ΔFosB in Tg2576 mice fed the low choline diet reflects more seizures than in the other two diets.

ΔFosB is a transcription factor that is linked to cognition. In the J20 mouse model of AD, elevated ΔFosB in GCs led to reduced cognition, and when ΔFosB was selectively reduced the cognition improved (Corbett et al. 2017 [↗](#)). Therefore, the reduction in ΔFosB by the high choline diet was important to show hyperexcitability was reduced and also important because it showed how high choline may benefit cognition.

Additional considerations

Although the low and high choline diets only had differences in choline, the intermediate diet had different amounts of choline as well as other constituents (**Supplementary Table 1** [↗](#)). Therefore, the only diet comparison with a difference restricted to choline is the low versus high choline diets (**Supplementary Table 1** [↗](#)). The intermediate diet was useful however, because numerous studies in AD mouse models employ this standard diet. Moreover, it may have promoted IIS as discussed above.

Although we failed to detect sex differences, there may actually have been differences if females had been analyzed according to their stage of the estrous cycle at death. This possibility is raised by prior data showing that rats and mice during diestrus have low excitability in hippocampal slices; at other cycle stages hyperexcitability occurs (Scharfman et al. 2003 [↗](#)).

The Tg2576 mouse model is one of many murine models of AD, and as with all models there are inherent limitations. The Tg2576 model of cerebral amyloid overexpression lacks tau pathology and recapitulates familial AD, whereas the majority of AD is sporadic. However, MCS has now demonstrated structural/functional benefits in several AD-relevant models, namely Tg2576 (this report), APP-PS1 (Mellott et al. 2017 [↗](#)), and Ts65Dn (Powers et al. 2016 [↗](#), Strupp et al. 2016 [↗](#), Powers et al. 2017 [↗](#), Alldred et al. 2021 [↗](#), Alldred et al. 2023 [↗](#)).

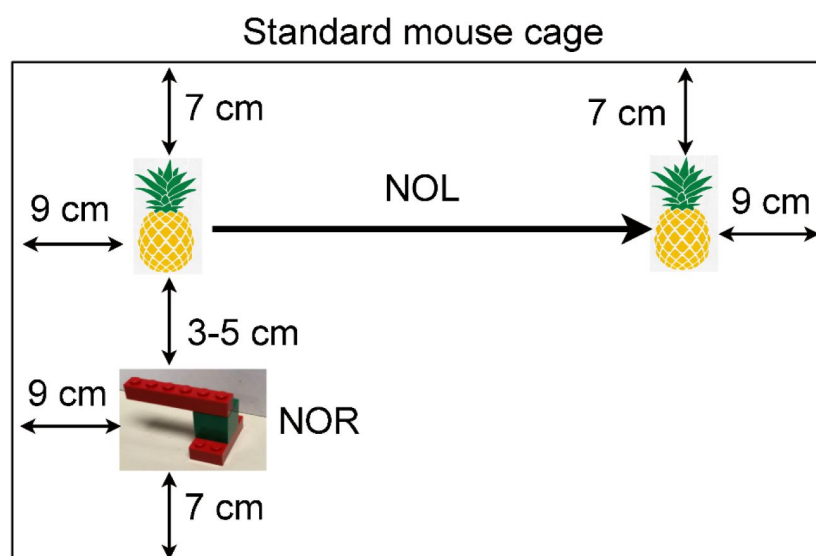
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	Low choline	Intermediate choline	High choline
Choline	1.1g/kg	2.0g/kg	5.0g/kg
Carbohydrate	690	570	690
Protein	177	236	177
Fat	120	170	120
Thiamin	0.020	0.015	0.020
Riboflavin	0.600	0.005	0.600
Pyridoxine	0.700	0.006	0.700
Niacin	3.000	0.780	3.000
Folic Acid	0.200	0.003	0.200
Biotin	0.020	0.002	0.020

Supplemental Table 1.

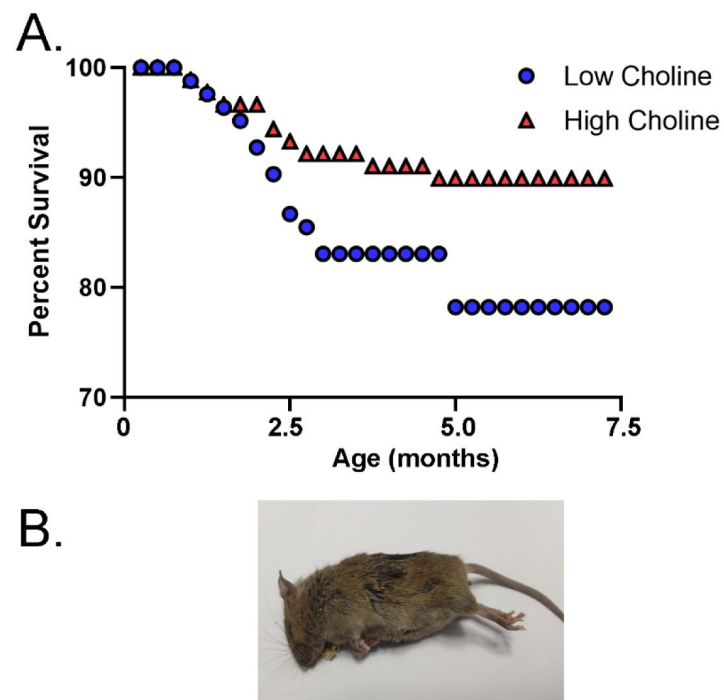
Comparison of the three diets used in this study.



Supplemental Figure 1.

Details of behavioral tasks.

A standard mouse cage is diagrammed with the locations of objects inside it. For NOL, there were two pineapple-like objects and one was moved across the cage for the test session. For the NOR test session, one pineapple-like object was removed and an object made of LEGO was placed in the location where the original pineapple-like object had been.



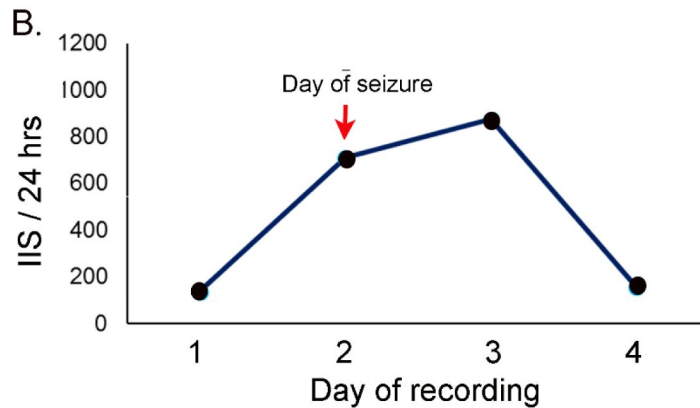
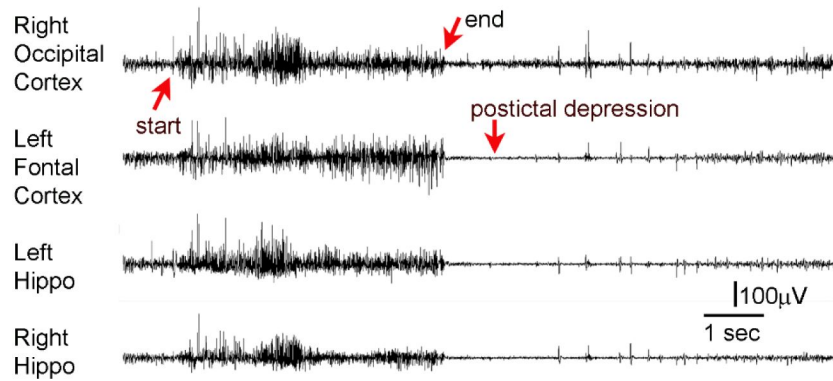
Supplemental Figure 2.

Mortality was high in mice treated with the low choline diet.

A. Survival curves are shown for mice fed the low choline diet and mice fed the high choline diet.

B. A photo of a mouse after sudden unexplained death. The mouse was found in a posture consistent with death during a convulsive seizure.

A. Seizure

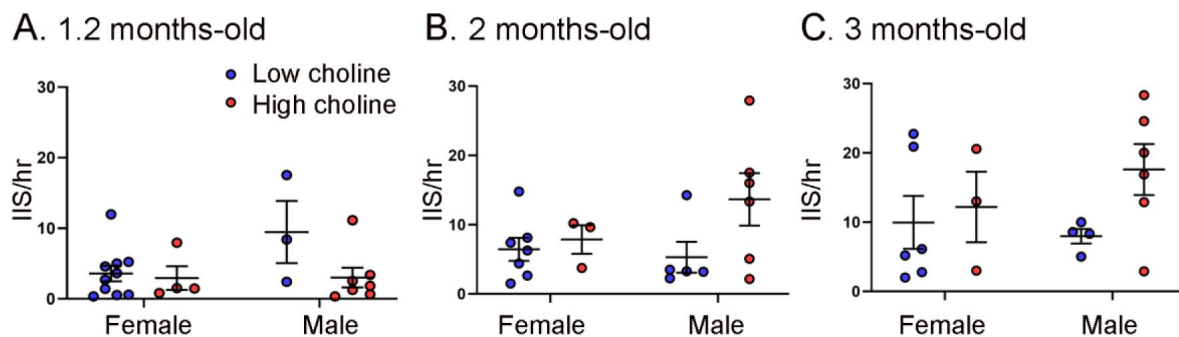


Supplemental Figure 3.

IIS frequency before and after seizures.

A. Representative EEG trace of a seizure in a 5 months-old Tg2576 mouse.

B. IIS frequency was quantified from continuous video-EEG for 4 days. During the first 24 hrs, IIS frequency was low. The next day a seizure occurred (arrow), and IIS frequency increased approximately 5 fold, and for the next 24 hr period. On day 4, IIS frequency returned to lower levels.



Supplemental Figure 4.

IIS frequency was similar for each sex.

A. At 1.2 months-old, female and male Tg2576 mice were not significantly different in IIS frequency for the low and high choline diets.

B-C. There were no significant sex differences at 2 (B) and 3 (C) months of age.

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Reviewer #1 (Public Review):

Summary:

Chartampila et al. describe the effect of early-life choline supplementation on cognitive functions and epileptic activity in a mouse model of Alzheimer's disease. The cognitive abilities are assessed by the novel object recognition test and the novel object location test, performed in the same cohort of mice at 3 months and 6 months of age. Neuronal loss was tested using NeuN immunoreactivity, and neuronal hyperexcitability was examined using $\square$ FosB and video-EEG recordings.

Strengths:

The study was designed as a 6-month follow-up, with repeated behavioral and EEG measurements through disease development, providing valuable and interesting findings on AD progression and the effect of early-life choline supplantation. Moreover, the behavioral data that suggest an adverse effect of low choline in WT mice are interesting and important beyond the context of AD.

Weaknesses:

1. The multiple headings and subheadings, focusing on the experimental method rather than the narrative, reduce the readability.
2. Quantification of NeuN and FosB in WT littermates is needed to demonstrate rescue of neuronal death and hyperexcitability by high choline supplementation and also to gain further insights into the adverse effect of low choline on the performance of WT mice in the behavioral test.
3. Quantification of the discrimination ratio of the novel object and novel location tests can facilitate the comparison between the different genotypes and diets.
4. The longitudinal analyses enable the performance of multi-level correlations between the discrimination ratio in NOR and NOL, NeuN and Fos levels, multiple EEG parameters, and premature death. Such analysis can potentially identify biomarkers associated with AD progression. These can be interesting in different choline supplementation, but also in the standard choline diet.

- <https://doi.org/10.7554/eLife.89889.1.sa1>

Reviewer #2 (Public Review):

Summary:

The authors demonstrated that maternal choline supplementation (MCS) improved spatial memory, reduced a marker of hyperexcitability/epilepsy ($\square$ FosB expression), and reduced oxidative stress (as measured by restored NeuN expression) in an Alzheimer's disease mouse model. This multidisciplinary study spanned behavior, EEG, and histological measures and constituted a large amount of work. Overall, the results supported that MCS does have important effects on hippocampal function, which may substantially impact human AD.

Strengths:

The strength of the group was the ability to monitor the incidence of interictal spikes (IIS) over the course of 1.2-6 months in the Tg2576 Alzheimer's disease model, combined with meaningful behavioral and histological measures. The authors were able to demonstrate MCS had protective effects in Tg2576 mice, which was particularly convincing in the hippocampal novel object location task.

Weaknesses:

Although choline deficiency was associated with impaired learning and elevated $\square$ FosB expression, consistent with increased hyperexcitability, IIS was reduced with both low and high choline diets. Although not necessarily a weakness, it complicates the interpretation and requires further evaluation.

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